STAT340 Lecture 08: Prediction

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Introduction

In these notes, we will introduce the task of prediction and begin talking about some of the fundamental tools for performing prediction, focusing on linear regression.

The task of prediction will be a theme for much of the remainder of the course, and it is not an exaggeration to say that prediction is the fundamental task that lies at the heart of machine learning.

Tasks like image classification (e.g., "does this image contain a cat or not?") are very naturally cast as prediction problems, as are many of the most basic problems in machine learning (e.g., predict how likely a person is to engage with the next piece of content in their feed).

Learning objectives

After this lesson, you will be able to

- Explain both simple and multiple linear regression
- Use R to run linear regression on a given data set and interpret the resulting coefficient estimates
- Explain what it means to associate a p-value to an estimated coefficient

Prediction: an overview

In a prediction problem, we are given data pairs $(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$ and we want to use X_i to predict Y_i.

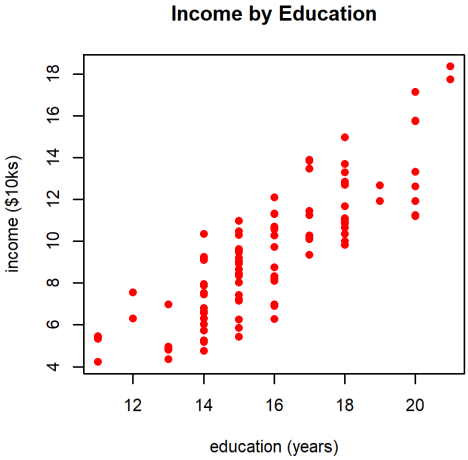
We call the X_i values the *predictors* (also called the *independent variables*), and we call our Y_i values the *responses* (or the *dependent variables* or the *outcomes*).

Example - education and income

Here, our (X_i, Y_i) pairs correspond to years of education (X_i) and income (Y_i). That is, our predictors are years of education, and our responses are income.

Our goal is to use this data to learn a function that maps years of education to income. That is, we want a function that takes years of education as input and outputs a prediction as to how much income we predict for a person with that income.

Linear regression makes a particular choice: we will learn a linear function that maps our predictors to (predicted) responses.



Review - Simple Linear Regression I

You discussed this problem at some length in STAT240, in the setting where our X_i and Y_i variables were all real-valued. That is, you learned about *simple linear regression*, in which we model our data as

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i,$$

where β_0 is the *intercept term*, β_1 is the coefficient associated with our predictors, and ϵ_i is an *error term*.

One way to think of this is that we imagine that given a value for the predictor X_i, say $X_i = x_i$, the "true" value of Y_i would be $\beta_0 + \beta_1 x_i$. However, we don't see this quantity exactly. Instead, due either to uncertainty in our data collection or to random factors (or both), we see a noisy version of this quantity, namely $Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$.

Review - Simple Linear Regression II

Example: income and education

We might predict that a person with X_i years of education makes

$$Y_i = -8.5 + 1.1X_i$$

thousands of dollars in salary. But there are lots of other factors in addition to years of education that might influence how much money someone makes. Including the error ϵ_i is a way to account for these unmeasured variables.

Review - Simple Linear Regression III

Example: astronomy

In the early days of astronomy, scientists like Johannes Kepler and Tycho Brahe were trying to fit the trajectories of planets in the sky to curves. When an astronomer takes a measurement of where a planet is, there are many sources of noise that cause them to make the measurement imprecisely. These sources of noise include both human error (e.g., imprecision in positioning the telescope, misreading a sight) and external sources (e.g., variations in humidity in the atmosphere change the way light bends as it reaches the observer's eye). If we model the position of a planet at time X_i as

$$Y_i = \beta_0 + \beta_1 X_i$$

for some choice of coefficients β_0 and β_1, we might account for these sources of *measurement error* via our error term ϵ_i. Of course, this linear model is incorrect– the planets follow quite complicated trajectories in the night sky, not simple straight lines. We'll come back to the matter of when linear models are or are not appropriate.

Writing down the model I

To recap, a simple linear model has an outcome y and single predictor x.

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_i + \epsilon_i \\ &= E(Y|X = x_i) + \epsilon_i \end{aligned}$$

where $i = 1, 2, \ldots, n$, and the error terms $\epsilon_i \sim N(0, \sigma^2)$ are independent over $i = 1, 2, \ldots, n$. We could write $\hat{y}_i = E(Y|X = x)$ notation is to stress that once we have chosen values for the coefficients β_0 and β_1, our prediction of the response of the i-th data point is $\hat{y}_i = \beta_0 + \beta_1 x_i$.

This equation represents our *model*, not the truth! We want to choose β_0 and β_1 so that this model describes our observed data as well as possible, but we have to bear in mind that this linearity assumption, that y_i is (exactly or approximately) expressible as $\beta_0 + \beta_1 x_i$, is an *assumption*. Our model will be good at predicting outcomes only in so far as this model agrees with reality.

Writing down the model II

The subscript i in our regression equation indexes the n observations in the dataset. Think of i as a row number. So another way to think about our model

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i,$$

is as a system of n equations,

$$\begin{aligned}
y_1 &= \beta_0 + \beta_1 x_1 + \epsilon_1 \\
y_2 &= \beta_0 + \beta_1 x_2 + \epsilon_2 \\
&\vdots \\
y_i &= \beta_0 + \beta_1 x_i + \epsilon_i \\
&\vdots \\
y_{n-1} &= \beta_0 + \beta_1 x_{n-1} + \epsilon_{n-1} \\
y_n &= \beta_0 + \beta_1 x_n + \epsilon_n.
\end{aligned}$$

Writing down the model III

The error terms ϵ_i in a linear model correspond, essentially, to the part of the variation in the data that remains unexplained by the deterministic portion of the model (encoded in the linear function $\beta_0 + \beta_1 x$).

One of the key assumptions of a linear model is that the residuals are independent and have mean zero, $\mathbb{E}\epsilon_i = 0$. Most typically, we further assume that they are normally distributed with mean 0 and variance σ^2. We'll do that here in these notes, but this choice can sometimes be "relaxed", in the sense that we may not need to assume that the errors are normal for linear regression to work, depending on what we want to do downstream. Later in your studies, when you take your mathematical statistics course, you'll put that statement on firmer ground; for now, we'll have to leave it vague.)

Interpreting simple linear regression I

So let's suppose that we're using the linear model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i.$$

It's useful to think about the expected value of Y given a particular value x: By the rules of expectation:

$$E(Y|X = x) = E(\beta_0 + \beta_1 x + \epsilon_i) = E(\beta_0) + E(\beta_1 x) + E(\epsilon_i)$$

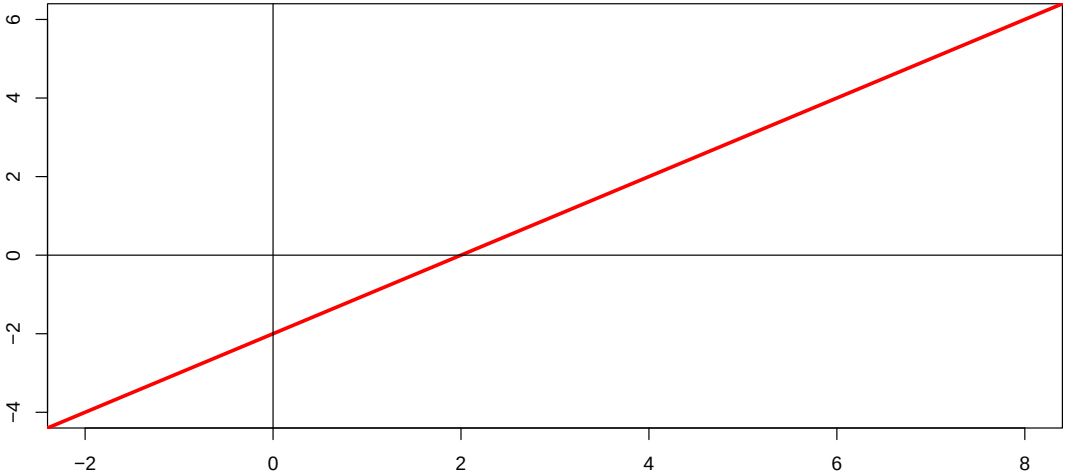
Because β_0 and $\beta_1 x$ are constant, and ϵ has a zero mean, this simplifies to

$$\hat{y} = E(Y|X = x) = \beta_0 + \beta_1 x.$$

Now, let's first consider what happens when $x = 0$. Then $\hat{y} = \beta_0$. Said another way, if we plotted the line $y = \beta_0 + \beta_1 x$, β_0 would be the intercept of our model.

Interpreting simple linear regression II

For example, here's the function with $\beta_0 = -2$ and $\beta_1 = 1$.



Interpreting simple linear regression III

The "typical" interpretation of the parameter β_0 is as describing what would happen, on average, if we observed a data point for which our predictor x were equal to zero.

β_1 is the slope. But how do we interpret it? Well, let's consider the avearge y value when X takes a particular value x.

$$y = E(Y|X = x) = \beta_0 + \beta_1 x.$$

What do we expect (predict in expectation) if X increases by 1 unit?

$$y' = E(Y|X = x + 1) = \beta_0 + \beta_1(x + 1).$$

The difference is

$$y' - y = \beta_0 + \beta_1(x + 1) - (\beta_0 + \beta_1 x) = \beta_1.$$

β_1 is the change in response that our model **predicts** if see a one unit increase in our predictor.

Interpreting simple linear regression IV

Example: income and education

Let's come back to our model predicting income (in tens thousands of dollars) from education, and suppose that we have fit a model of the form

$$y = -8.5 + 1.1x.$$

So our coefficients are $\beta_0 = -8.5$ and $\beta_1 = 1.1$. Thus, our model predicts that an increase in education by 1 year is associated with in an increase of \$11K in salary (1.1 times our unit of measurement, \$10K/year).

Similarly, since $\beta_0 = -8.5$, our model "predicts" that a person with zero years of education will receive a salary of -\$85K per year.

Interpreting simple linear regression V

Example: a cautionary tale

Interpreting the intercept as describing the response at $x = 0$ often doesn't make sense.

Suppose that we fit a model that predicts income (in thousands of dollars) from height (in centimeters),

$$y = 10 + 0.4x,$$

where x is height in centimeters.

So our model "predicts" that a person with height $x = 0$ would make a salary of \$10,000 per year. This is a prediction on an input that we don't really every expect to encounter in the real word.

Danger of Extrapolation

Our linear regression model really only makes sense over a **certain range of values**, and we should be careful when using our model to extrapolate to "strange" values of x.

Even though we might be able to associate a response with any particular input x, that doesn't mean that every such input is realistic.

Caution: causality

Having fit a model it is tempting to say that "increasing the value of the predictor by one unit causes an increase in the response by (so many) units."

Still, we should be careful to avoid giving a causal interpretation to our findings.

For example, suppose that we fit a linear regression model to predict the number of cancer cases per year in Wisconsin based on pollution levels (measured in, say, PM2.5), and we estimate $\beta_1 = 10.2$. We might be tempted to say that a unit increase of PM2.5 *causes, on average*, an additional 10.2 cancer cases.

This is a stronger statement than what we can conclude from a linear model fitted in this way. We can only say that a unit increase in PM2.5 is *associated with* an increase of 10.2 cancer cases. This is the old "correlation is not causation" saying, wearing a slightly different hat.

There is a whole area in statistics called *causal inference* that attempts to use statistics to make causal statements, but it is, unfortunately, outside the scope of the course.

Fitting the model

Suppose that we have chosen values of our coefficients β_0 and β_1 in our regression model. The estimated coefficients will be denoted $\hat{\beta}_0$ and $\hat{\beta}_1$

How do we decide how "good" or "bad" this choice of coefficients is? We need a function that takes a particular choice of $\hat{\beta}_0$ and $\hat{\beta}_1$ and outputs a number that measures how well or poorly the resulting model describes our data.

In the setting where larger values of this function correspond to worse model fit, we call this kind of a function a **loss function**: it takes a choice of model parameters (i.e., coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$, in our case), and outputs a number that measures how poorly our model fits the data.

Choosing a loss: sum of squares

There are lots of functions we could choose to use as our loss function, but by far the most common choice is the **residual sum of squares** (RSS), sometimes called the **sum of squared errors** (SSE):

$$\ell\left(\hat{\beta}_0, \hat{\beta}_1\right) = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} \left(y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right)\right)^2.$$

The terms $y_i - \hat{y}_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)$ are called the *residuals*.

The residuals are what is left over after we try to predict the responses from the predictors $x_1, x_2, \ldots, x_n$.

Our goal is to choose our coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize the sum of squared residuals loss in the equation above. We call this *ordinary least squares* (OLS) regression.

Minimizing the loss

If we take partial derivatives we can solve for the optimal choices of $\hat{\beta}_0$ and $\hat{\beta}_1$.

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$
$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2},$$

where $\bar{x}$ and $\bar{y}$ are the means of the predictors and responses, respectively:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \quad \text{and} \quad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i.$$

Interpreting the estimated slope

By our definition of $\hat{\beta}_1$, we have

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

By introducing $\frac{1}{n-1}$ into the numerator and denominator we can rewrite this as

$$\hat{\beta}_1 = \frac{s_y}{s_x}\hat{\rho}_{x,y}.$$

In other words, the slope of our model is the ratio of the standard deviations, scaled by the correlation between our predictors and responses.

If your data has been standardized (i.e. $s_y = s_x = 1$) then $\hat{\beta}_1$ is simply the sample correlation.

Interpreting the intercept term $\hat{\beta}_0$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Bearing this in mind, what would we predict the value of y to be when $x = \bar{x}$?

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

But we can replace $\hat{\beta}_0$ in this equation

$$\hat{y} = (\bar{y} - \hat{\beta}_1 \bar{x}) + \hat{\beta}_1 \bar{x} = \bar{y}$$

Also note that if you have standardized your data (hence $\bar{x} = \bar{y} = 0$) then $\hat{\beta}_0 = 0$.

Variance of estimates

After fitting, we can find our predicted $\hat{y}_i$, i.e. the y values on the line.

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

as well as our model residuals $\hat{\epsilon}_i$

$$\hat{\epsilon}_i = y_i - \hat{y}_i$$

We also get an estimate of the variance of the residuals σ^2, which happens to be very useful in computing other statistics. The reason is that **the larger the residuals' variance, the less precisely we can estimate our regression coefficients**, which should make a lot of sense.

$$\hat{\sigma}^2 = \text{mean squared error} = \frac{SSE}{n-2} = \frac{1}{n-2}\sum_i (y_i - \hat{y}_i)^2$$

The $n-2$ in the denominator is because we have 2 estimated parameters in our model.

Simple linear regression: Lead example I

In the 1920s, Thomas Midgley Jr. discovered that adding tetraethyllead to gasoline decreased engine knocking (i.e., when fuel doesn't fully ignite in an engine cylinder, which may damage the engine). He won the 1923 Nichols medal, a prestigious prize in chemistry, for his discovery.

The result of burning tetraethyllead in gasoline resulted in high levels of lead in the atmostphere. By the 1950s to 70s, researchers started to suspect that increased lead levels in the atmosphere was causing widespread lead poisoning, with symptoms ranging from depression, loss of appetite, and amnesia to anemia, insomnia, slurred speech, and cognitive impairment. Starting in the 1980s, the use of tetraethyllead in gasoline started to be phased out.

In more recent years, a more controversial theory has emerged, suggesting that exposure to lead (be it in the atmosphere or in paint in older buildings) correlates with incidents of violent crime later in life.

Simple linear regression: Lead example II

Let's look at a dataset that contains atmospheric lead content levels and aggravated assault rates for several cities in the US and see if we can build a simple linear regression model to explain the trend and make predictions.

```
lead <- read.csv('lead.csv')
# First things first: let's look at the data.
head(lead)
##      city air.pb.metric.tons aggr.assault.per.million
## 1 Atlanta                421                     1029
## 2 Atlanta                429                      937
## 3 Atlanta                444                      887
## 4 Atlanta                457                      533
## 5 Atlanta                461                     1012
## 6 Atlanta                454                      848
```

The variables we are interested in are lead levels in the atmosphere (measured in metric tons of lead emitted) and the aggravated assault rate per million 22 years later.

Simple linear regression: Lead example III

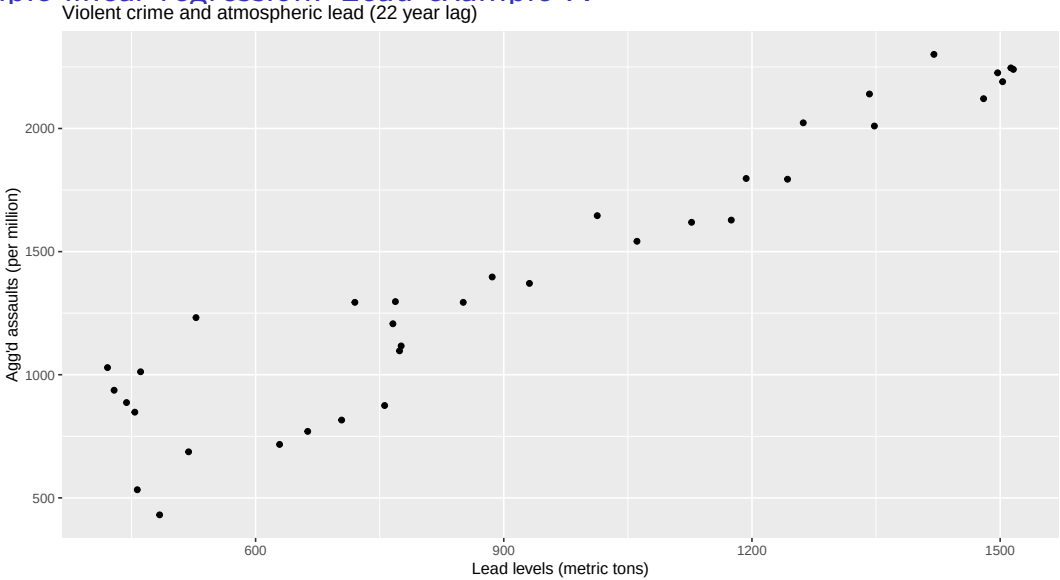
We want to predict the assault rate from lead levels, so our predictor (or explanatory variable or independent variable, if you prefer) is lead levels, and our response (or dependent variable) is the assault rate.

For simplicity, let's focus on the city of Atlanta.

```
atlanta_lead <- lead[ lead$city=='Atlanta', ];
## Alternative approach, using filter in dplyr:
# library(dplyr)
# atlanta_lead  lead %>% filter(city == "Atlanta")

# Plot the data to get a look at what's going on.
library(ggplot2)
pp <- ggplot( atlanta_lead,
          aes(x=air.pb.metric.tons, y=aggr.assault.per.million));
pp <- pp + geom_point();
pp <- pp + labs( x="Lead levels (metric tons)",
        y="Agg'd assaults (per million)",
       title="Violent crime and atmospheric lead (22 year lag)" )
```

Simple linear regression: Lead example IV



Simple linear regression: Lead example V

Visually, it's quite clear that assaults are well-predicted as a linear function of lead levels. One thing that is already perhaps of concern is that the data appears to be a bit more "spread out" in the vertical direction for lower lead levels. This is a bit concerning in light of our assumption that the error terms were all distributed according to a normal with mean 0 and variance σ^2.

To fit a linear model in R, we use the `lm()` function.

The syntax is `lm(y ~ 1 + x, data=dframe)`, where `dframe` is the data frame containing our data, and `y ~ 1 + x` means to regress the variable `y` (i.e., the column `y` in the dataframe `dframe`) against the variable `x` and an intercept term (that's the `1` in the model formula):

$$y = \beta_0 + \beta_1 x + \epsilon$$

Note that the `1` in the model formula `y ~ 1 + x` is completely optional– R will include an intercept term automatically.

Simple linear regression: Lead example VI

The function `lm` returns an object of the class `lm`. This is an object that contains a bunch of information about our fitted model.

So in our case, we want to regress `aggr.assault.per.million` against `air.pb.metric.tons`. That is, we want a model like

$$\text{agg.assault} = \beta_0 + \beta_1 \cdot \text{air.pb}$$

So we'll write that as `aggr.assault.per.million ~ 1+ air.pb.metric.tons`. Let's try fitting the model and then we'll ask R to summarize our model. Running `summary()` on the model object gives us a variety of useful summary statistics and other information about the fitted model.

```
atlanta_lead_lm <- lm(aggr.assault.per.million ~ 1 + air.pb.metric.tons,
                      data=atlanta_lead)
```

Simple linear regression: Lead example VII

```
summary(atlanta_lead_lm)
##
## Call:
## lm(formula = aggr.assault.per.million ~ 1 + air.pb.metric.tons,
##     data = atlanta_lead)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -356.36  -84.55    6.89  122.93  382.88
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)        107.94276   80.46409   1.342    0.189
## air.pb.metric.tons   1.40375    0.08112  17.305   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 180.6 on 34 degrees of freedom
## Multiple R-squared:  0.898,  Adjusted R-squared:  0.895
## F-statistic: 299.4 on 1 and 34 DF,  p-value: < 2.2e-16
```

Simple linear regression: Lead example VIII

For now, let's notice in particular that we have estimated coefficients, accessible in the `coefficients` attribute of our model object:

```
atlanta_lead_lm$coefficients
##        (Intercept) air.pb.metric.tons
##         107.942757           1.403746
```

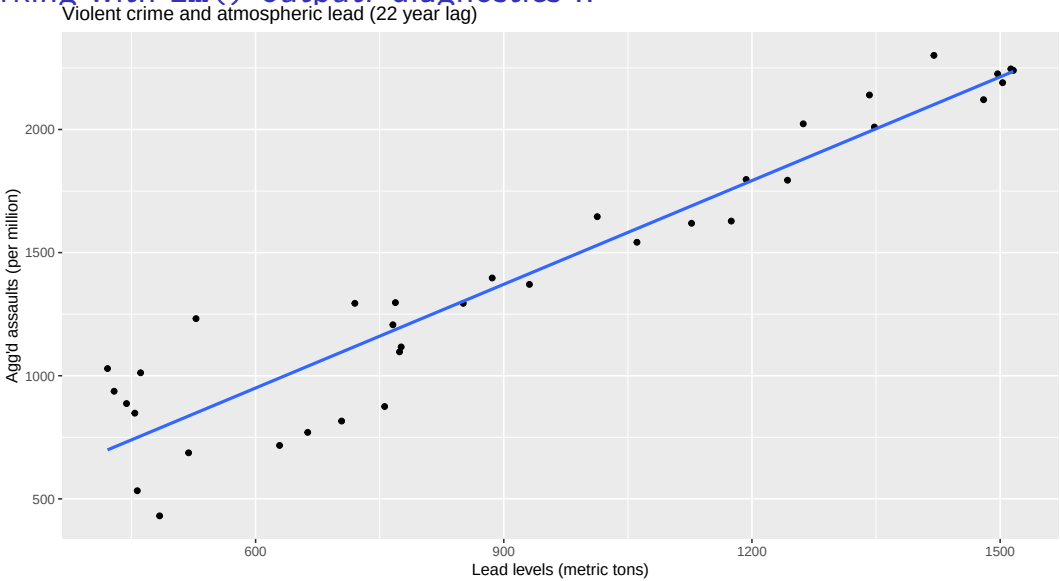
So our model predicts that in Atlanta, an increase of one metric ton of lead in the atmosphere is associated with an increase of about 1.4 aggravated assaults per million people. Similarly, the intercept indicates that our model predicts that in the absence of any lead in the atmosphere, there would be about 108 aggravated assaults per million people.

Working with `lm()` output: diagnostics I

Let's look at how our fitted model tracks the data by overlaying the fitted line on our scatterplot above. One way to do this would be to use `abline` with the entries of `atlanta_lead_lm$coefficients` to specify the slope and intercept, but `lm` is so common that `ggplot2` has this same basic functionality built-in, in the form of `geom_smooth()`.

```
pp <- ggplot( atlanta_lead,
              aes(x=air.pb.metric.tons,y=aggr.assault.per.million));
# The argument `se` specifies whether or not to include a
# confidence interval around the plotted line.
# We'll talk about that later.
# For now we'll just suppress the CI with se=FALSE
pp <- pp +geom_point() + geom_smooth(method="lm",
                                     formula="y~x",
                                     se=FALSE);
pp <- pp + labs( x="Lead levels (metric tons)",
            y="Agg'd assaults (per million)",
            title="Violent crime and atmospheric lead (22 year lag)" )
pp
```

Working with `lm()` output: diagnostics II



Working with `lm()` output: diagnostics III

Looks like a pretty good fit! Let's look at some of the other information included in the output of `lm()`. The `fitted` function gets fitted y-values (i.e., points on line of best fit)

```
fitted(atlanta_lead_lm)
##         1         2         3         4         5         6         7         8
##  698.9200  710.1499  731.2061  749.4548  755.0698  745.2436  787.3560  836.4871
##         9        10        11        12        13        14        15        16
##  849.1208  990.8992 1038.6266 1096.1802 1118.6401 1169.1750 1183.2125 1187.4237
##        17        18        19        20        21        22        23        24
## 1194.4424 1197.2499 1302.5309 1351.6620 1414.8306 1529.9378 1597.3176 1689.9649
##        25        26        27        28        29        30        31        32
## 1757.3447 1782.6122 1852.7995 1879.4707 1991.7704 2000.1929 2101.2626 2185.4874
##        33        34        35        36
## 2209.3511 2231.8110 2217.7735 2236.0222
```

Working with `lm()` output: diagnostics IV

The `residuals` function gets you the residuals (the difference between the observed response y and the response predicted by our model)

```
residuals(atlanta_lead_lm)  # We can also use resid()
##           1           2           3           4           5           6
##  330.080025  226.850054  155.793859 -216.454844  256.930171  102.756395
##           7           8           9          10          11          12
## -356.355996 -149.487118  382.879164 -273.899218 -268.626594 -280.180195
##          13          14          15          16          17          18
##  175.359863 -294.175006   23.787530  109.576291  -97.442440  -80.249933
##          19          20          21          22          23          24
##   -8.530910   45.337967  -43.830619  116.062179  -55.317646  -70.964906
##          25          26          27          28          29          30
## -129.344731   14.387834  -58.799484  143.529335  148.229626    9.807148
##          31          32          33          34          35          36
##  199.737410  -64.487372   16.648940   14.188998  -27.773538    3.977759
```

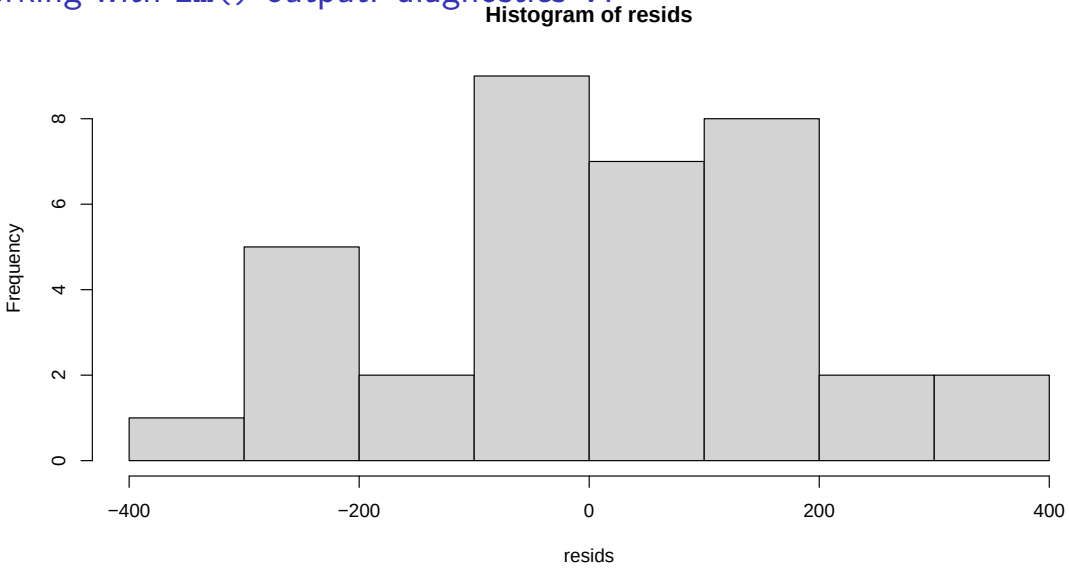
Working with `lm()` output: diagnostics V

These residuals reflect the error between our model's prediction and the true observations, and they are often quite informative.

Recall that our model assumes that the observation errors ϵ_i are normally distributed about zero, with a shared variance σ^2. To check that this assumption is (approximately) true, we can plot the residuals:

```
resids <- residuals(atlanta_lead_lm)
hist(resids)
```

Working with `lm()` output: diagnostics VI



Homoscedasticity I

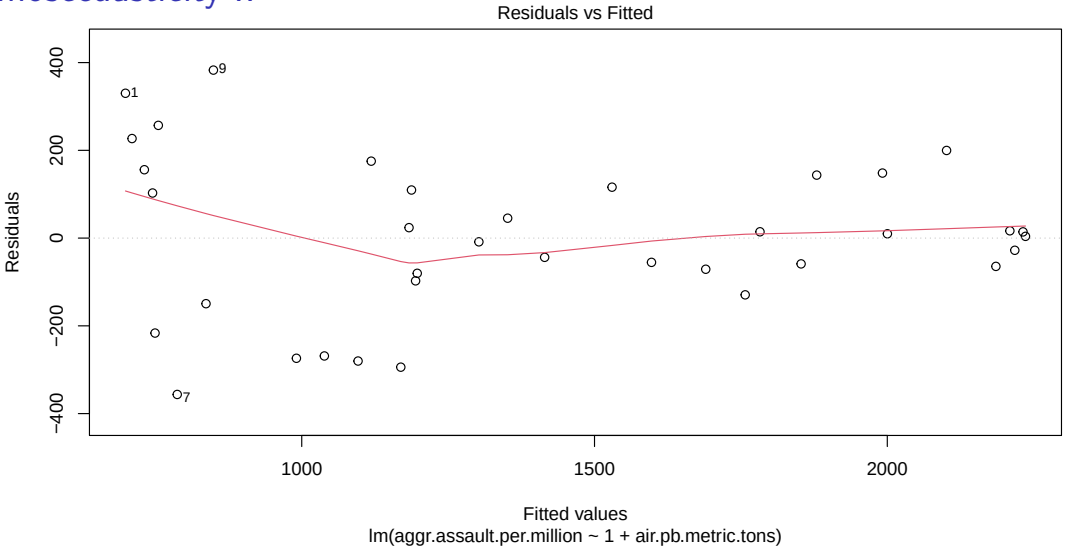
Another important point, far more important that the normality assumption, is that the variance of the errors ϵ_i does not depend on the predictor X_i. This is referred to as "homogeneity of variance", more commonly called homoscedasticity. Its absence, *heteroscedasticity*, wherein the variance of the error terms varies with X_i, can be a big problem for linear regression.

So let's check for it, just visually for now. We want to plot the residuals as a function of their predictor values. If our errors are homoscedastic, we should observe the variance of the residuals about the horizontal line $y = 0$ to be more or less constant along the x-axis. R will do this for us automatically if we call `plot` on our model object. In fact, R will make several plots for us automatically, and return those plots in a list-like object.

The residuals as a function of the `x` values is the first of these, and we can access it with the `which` keyword to `plot()`.

```
plot(atlanta_lead_lm, which=1)
```

Homoscedasticity II



Homoscedasticity III

The red line is a moving average; if our residuals satisfy the zero-mean assumption this line should be horizontal.

Inspecting this plot, it looks as we suspected– the residuals for smaller lead atmospheric levels have slightly higher variance, and tend to be biased toward positive values.

Still, this doesn't look especially extreme.

Assessing normality of the residuals I

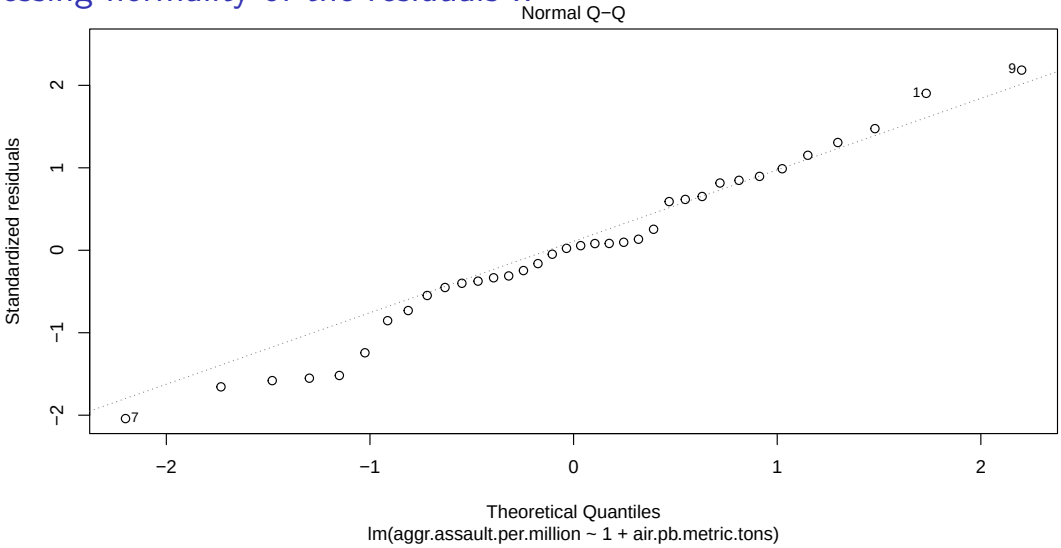
The stronger assumption is that the errors are normal with mean zero.

We saw that their histogram above looked pretty reasonable.

A better check for fit is to construct a Q-Q plot.

```
plot(atlanta_lead_lm, which=2)
```

Assessing normality of the residuals II



Assessing normality of the residuals III

A Q-Q-plot displays the quantiles (i.e., the percentiles) of our observed data against the quantiles of the normal distribution.

If our data were perfectly normally distributed, then the Q-Q plot would look like a straight line with slope 1.

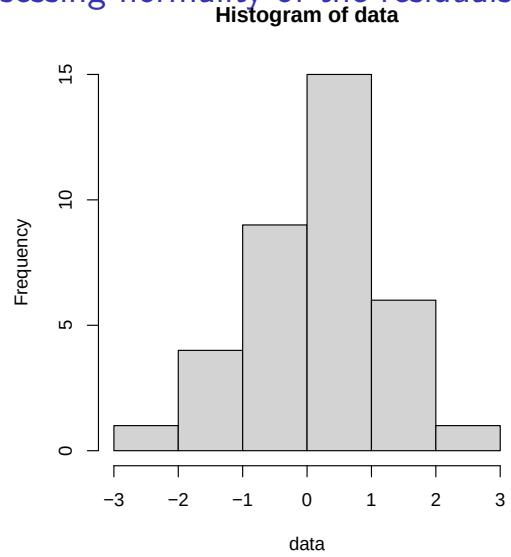
If our data is not, the Q-Q plot will look far different.

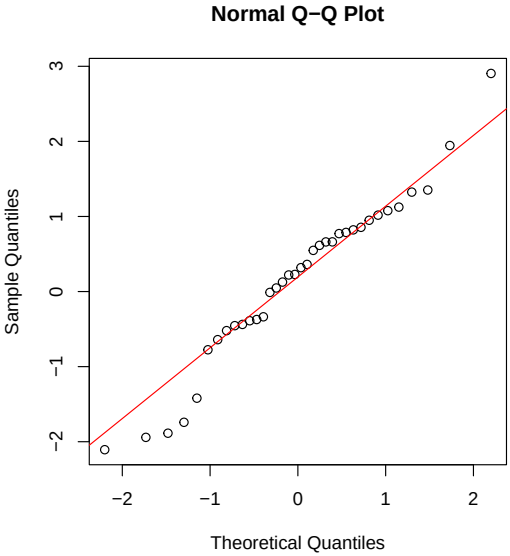
Just to see an example of this, let's generate some data from a t-distribution (which *looks* normal, but has "heavier tails"), and look at the Q-Q plot.

```r
data <- rt(n=36, df=3, ncp=0);
par(mfrow=c(1,2))
hist(data)

qqnorm(data);
# Add a line to the plot to indicate the behavior we would
# expect to see if the data were normal.
qqline(data, col='red');
```

Assessing normality of the residuals IV



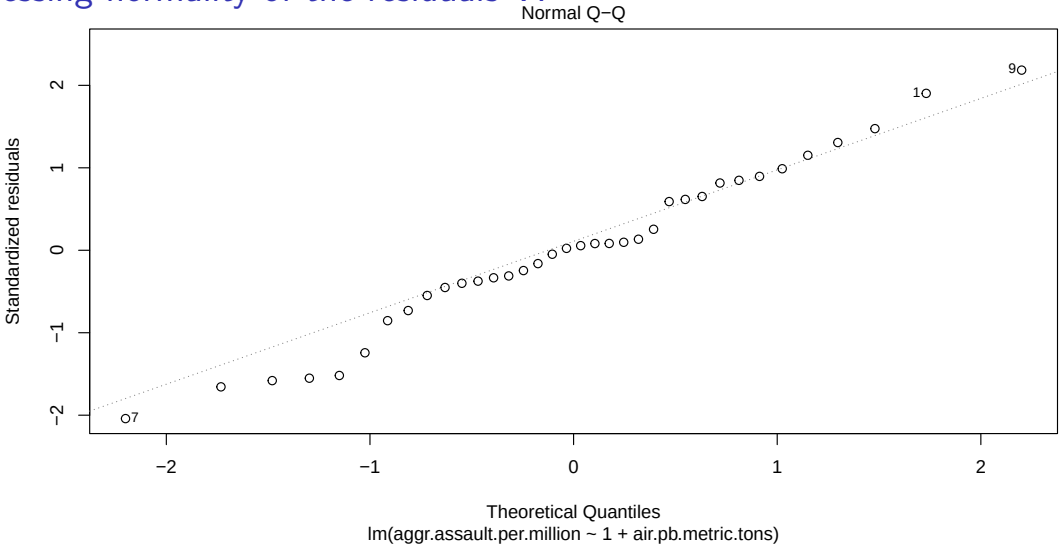


Assessing normality of the residuals V

Here's our lead level data again.

```r
# We could also call qqnorm(atlanta_lead_lm$residuals)
plot(atlanta_lead_lm, which=2)
```

Assessing normality of the residuals VI



Assessing normality of the residuals VII

There are some slightly concerning spots there, especially in the bottom-left, but it's not too extreme.

The best way to develop this intuition is to generate data from a normal distribution and see what types of qq plots are typical.

Testing and confidence intervals for coefficients I

So we've established that our model is a reasonably good fit for the lead data, at least in the sense that the trend in the data follows our plotted line and such.

Can we conclude from this that the association between lead and aggravated assault rate is "real"? It's possible, after all, that the observed association is merely due to chance.

Well, in our discussions of hypothesis testing we saw a number of tools for checking if observations were merely due to chance or not. Linear regression has its own set of tools for checking whether observed coefficient estimates are "merely due to chance".

Let's look back at our model summary and let's pay particular attention to the "coefficients" part of the output.

Testing and confidence intervals for coefficients II

```
summary(atlanta_lead_lm)
##
## Call:
## lm(formula = aggr.assault.per.million ~ 1 + air.pb.metric.tons,
##     data = atlanta_lead)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -356.36  -84.55    6.89  122.93  382.88
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)        107.94276   80.46409   1.342    0.189
## air.pb.metric.tons   1.40375    0.08112  17.305   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 180.6 on 34 degrees of freedom
## Multiple R-squared:  0.898,  Adjusted R-squared:  0.895
## F-statistic: 299.4 on 1 and 34 DF,  p-value: < 2.2e-16
```

Testing and confidence intervals for coefficients III

The `Coefficients` table includes our estimates for the coefficients, standard errors for those estimates, t-values (i.e., a test statistic) for those statistics and, at the far right of the table, a column headed `Pr[>|t|]`.

Notice that in this case, the *p*-value associated to our estimate of the coefficient of lead levels is quite small. That indicates that an estimate this extreme (or more extreme) is highly unlikely to have arisen entirely by chance.

Now, it is in dealing with these *p*-values that we need to be a bit careful about our model assumptions. If the assumption of normal, homoscedastic errors is violated, things can go wrong with these p-values. But since our Q-Q plot indicated that our residuals were reasonably normal-looking, we can be somewhat confident that this is reflecting a real effect (well, and it's pretty clear just from the plot that there's a linear relationship. . .).

Testing and confidence intervals for coefficients IV

This p-value arises, in essence, from a t-test. This t-test is designed to test the null hypothesis

$$H_0 : \beta_1 = 0.$$

In this case, our p-value associated with β_1 is quite small, indicating a correlation between lead levels and aggravated assault levels. It does **not** imply causation, though, as you well know by now. Nonetheless, this seems to be fairly convincing evidence that there is an association.

Testing and confidence intervals for coefficients V

Thinking back to our brief discussion of the connection between confidence intervals and testing, you won't be surprised to learn that we can also compute confidence intervals for the true value of the coefficients.

```
confint(atlanta_lead_lm, level=0.95)
##                          2.5 %     97.5 %
## (Intercept)        -55.579958 271.465472
## air.pb.metric.tons   1.238891   1.568602
```

Of course, we also tested the hypothesis $H_0 : \beta_0 = 0$, and the p-value is not especially small (also reflected in the fact that the confidence interval includes 0). This indicates that our intercept term was not statistically significantly different from zero.

Note, however, that just because our p-value associated to the intercept term isn't especially small, that doesn't mean that the intercept isn't useful for prediction.

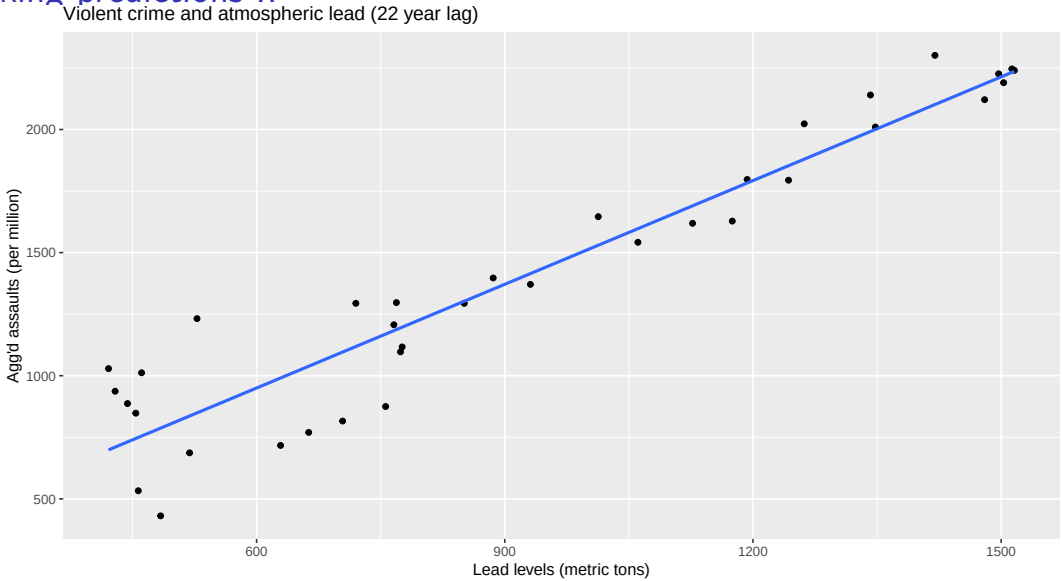
Making predictions I

Suppose that tomorrow a chemical company near Atlanta has an incident, and lead is released into the atmosphere. Suppose that the new atmospheric levels of lead are found to be 1300 metric tons. What would you predict the approximate aggravated assault rate to be 22 years later?

Well, let's start by just looking at a plot of our model again.

```
pp <- ggplot( atlanta_lead,
              aes(x=air.pb.metric.tons,y=aggr.assault.per.million));
# The argument `se` specifies whether or not to include a
# confidence interval around the plotted line.
# We'll talk about that later.
# For now we'll just suppress the CI with se=FALSE
pp <- pp +geom_point() + geom_smooth(method="lm",
                                     formula="y~x",
                                     se=FALSE);
pp <- pp + labs( x="Lead levels (metric tons)",
         y="Agg'd assaults (per million)",
         title="Violent crime and atmospheric lead (22 year lag)" )
```

Making predictions II



Making predictions III

Just looking at the plot, we see that at x-value 1300, our fitted line is just about 2000 aggravated assaults per million people.

R model objects (including linear regression) support a function called `predict`. We pass our model, and some data (i.e., x-values), and `predict()` outputs our model's predicted responses at those values.

```
predict(atlanta_lead_lm, newdata=data.frame(air.pb.metric.tons=1300))
##        1
## 1932.813
```

A "confidence interval" for the prediction estimates the **mean response value** for the given X. The variability comes from uncertainty about our parameter estimates.

```
predict(atlanta_lead_lm, newdata=data.frame(air.pb.metric.tons=1300), interval="confidence")
##        fit      lwr      upr
## 1 1932.813 1845.229 2020.397
```

Making predictions IV

A "prediction interval" estimates plausible values for **one particular** response value (as opposed to the mean). The interval is wider because we add on the variability from individual to individual.

```
predict(atlanta_lead_lm, newdata=data.frame(air.pb.metric.tons=1300), interval="predict")
##        fit      lwr      upr
## 1 1932.813 1555.393 2310.233
```

Making predictions V

Suppose the company continues to release more lead into the atmosphere, and next year, the levels are measured to be 2000 metric tons. Can we use our model to predict what aggravated assault rates might look like 22 years later?

2000 metric tons is rather far outside the range of our observed predictor values.

Predictions made far outside the range of observed predictors have to be treated carefully They may be reliable, but they also may not. The reliability of a prediction usually decreases the further away it is from your data.

Assessing model fit I

Once we've fit a model to the data, how do we tell if our model is good or not?

We fit our model to data by minimizing the sum of squares (we're sticking with simple linear regression here for simplicity– this idea extends to multiple linear regression in the obvious way),

$$\ell(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} \left(y_i - (\beta_0 + \beta_1 x_i)\right)^2 .$$

So what better way to measure how good our model is than using precisely this quantity?

Assessing model fit II

We define the residual sum of squares (RSS; also called the sum of squared errors, SSE) to be the sum of squared errors of our model. That is, letting $\hat{\beta}_0$ and $\hat{\beta}_1$ be our estimates of the coefficients,

$$\text{RSS} = \text{SSE} = \sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} \left(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i) \right)^2$$

If RSS is lower then the model fits "better'' but again this is hard to measure because RSS is not a standardized statistic.

We can compare this to the total sum of squares (TSS) - the sum of squared residuals for a model

$$\hat{y} = \bar{y}$$

$$\text{TSS} = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

Assessing model fit III

This leads to a standardized statistic, the **Coefficient of Determination**, or R-squared, which can be interpreted as measuring the proportion (between 0 and 1) of the variation in Y that is explained by the variation in X.

$$R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}},$$

Assessing model fit IV

In the case of simple linear regression, things simplify so that $R^2 = r^2$, where r is the correlation coefficient between X and Y,

When this quantity is close to 1, we can be confident that our linear model is accurately capturing a trend in the data.

We will see that a generalized definition is

$$R^2 = r^2_{y,\hat{y}}$$

Review I

- The simple linear regression model
- predictor / response variables
- assumptions of SLR model
- interpretation of SLR components
- correlation is not causation
- residuals
- loss function of OLS: Sum of squared residuals
- slope estimate formula and interpretation
- intercept estimate formula and interpretation
- units of β estimates
- mean squared error, estimate of σ_ϵ^2
- variance of β estimates
- OLS in R, and summary output
- OLS diagnostics (residual plot, residual QQ plot)
- inference and confidence intervals for coefficients
- predictions & prediction intervals
- R^2 and r